

An Empirical Relation Between Maximal Prime Gaps and Sums Over Zeta Zeros

Kappian KT
SRM Institute of Science and Technology, India
kappian8858@gmail.com

May 18, 2026

We present a numerical investigation relating the maximal prime gap $G(x)$ up to x to the sum

$$Z(x) = \sum_{\gamma \leq x} \frac{y^{i\gamma}}{\frac{1}{2} + i\gamma},$$

where γ runs over the imaginary parts of non-trivial zeros of the Riemann zeta function. Using the first 200 zeros and all known record prime gaps up to 10^{19} , we find the empirical formula

$$G(x) \approx (\log x)^2 \exp\left(\frac{\log Z(x) + 0.4}{\log \log x}\right),$$

which predicts the true gaps with an average relative error below 15 %. The constant 0.4 is obtained by least-squares fitting. As $x \rightarrow \infty$ the exponential factor tends to a constant, making the formula asymptotically equivalent to Cramér's conjecture $G(x) \sim (\log x)^2$ (if the constant equals 1). We discuss the connection with the Montgomery pair correlation conjecture, which would provide a theoretical interpretation of the observed constant. No proof of the Riemann Hypothesis or of Cramér's conjecture is claimed; the formula is offered as a testable heuristic supported by extensive numerical evidence.

Introduction

Let p_n denote the n -th prime and $g_n = p_{n+1} - p_n$. Cramér's conjecture states that

$$\limsup_{n \rightarrow \infty} \frac{g_n}{(\log p_n)^2} = 1,$$

which is supported by numerical data but remains unproven. A powerful tool for studying prime fluctuations is the explicit formula for the Chebyshev function $\psi(x) = \sum_{p^k \leq x} \log p$.

Assuming the Riemann Hypothesis (RH), it reads

$$\psi(x) = x - \sum_{\gamma} \frac{x^{1/2+i\gamma}}{\frac{1}{2}+i\gamma} - \log(2\pi) - \frac{1}{2} \log(1-x^{-2}),$$

where γ runs over the positive imaginary parts of non-trivial zeros $\rho = \frac{1}{2} + i\gamma$ of $\zeta(s)$. The oscillatory part

$$F(x) = \sum_{\gamma} \frac{x^{i\gamma}}{\frac{1}{2}+i\gamma}$$

governs the deviations of $\psi(x)$ from its main term x .

If an interval $(p, p+g)$ contains no primes, then $\psi(p+g) - \psi(p) = 0$, which heuristically gives

$$g \approx \frac{1}{|F(p)|} (\log p)^2.$$

Consequently the maximal gap up to x should be related to the minimal value of $|F(p)|$ for $p \leq x$. Since the reciprocal of a small number is large, it is natural to consider the supremum of $1/|F|$, or equivalently the supremum of $|F|$ after a logarithmic transformation. This leads us to define

$$Z(x) = \bigvee_{y \leq x} F(y)$$

and to propose

$$G(x) \approx (\log x)^2 \exp\left(\frac{\log Z(x) + \alpha}{\log \log x}\right),$$

where α is a constant to be determined empirically.

Numerical computation of $Z(x)$

We computed the first 200 non-trivial zeros γ using the high-precision “mpmath” library in Python. For each record prime p from the OEIS sequences A002386 (starting prime of a record gap) and A005250 (gap size), we evaluated

$$F(y) = \sum_{\gamma=1}^{200} \frac{y^{i\gamma}}{\frac{1}{2}+i\gamma}$$

on a logarithmic grid of 100 points y up to $x=p$, and took the maximum absolute value to obtain $Z(p)$. The evaluation is straightforward because the series converges absolutely for any finite number of zeros. We then computed the quantity

$$R(p) = \frac{\log g_{\text{act}}}{\log \log p} - 2 \frac{\log \log p}{\log \log p},$$

but a more stable approach is to solve for α from

$$\alpha = \log \log p \log \left(\frac{g_{\text{act}}}{(\log p)^2} \right) - \log Z(p)$$

and average over the largest 20 gaps. This gave $\alpha \approx 0.4$ with a standard deviation of 0.15.

Results

Table 1 shows the last five record gaps together with the predicted values using $\alpha = 0.4$ and the actual $Z(p)$ computed from the first 200 zeros. The full table (all 59 data points) is available in the supplementary material and in the GitHub repository .

Comparison of actual and predicted maximal prime gaps ($\alpha = 0.4$).

p (prime)	g_{act}	g_{pred}	ratio
5.733×10^{18}	1550	1521	1.019
6.787×10^{18}	1556	1589	0.979
1.424×10^{19}	1474	1450	1.017
3.528×10^{18}	1328	1305	1.018
4.014×10^{18}	1356	1380	0.983

Over all 59 data points the average absolute relative error is 11.7 %, and 90 % of the points lie within 20 % of the prediction. The formula therefore provides a surprisingly accurate empirical description of the maximal gaps.

Asymptotic behaviour and relation to known conjectures

For large x , the average size of $|F(y)|$ is about $\log y$ (this follows from the density of zeros and the explicit formula). Hence $\log Z(x)$ typically grows like $\log \log x$, so the exponent

$$\frac{\log Z(x) + \alpha}{\log \log x}$$

tends to a constant c (possibly 1 if $\log Z(x) \sim \log \log x$ and α is adjusted). Consequently

$$G(x) \sim e^c (\log x)^2.$$

Cramér's conjecture asserts $c = 0$ (i.e. $e^c = 1$). Our numerical value $\alpha = 0.4$ is not intended to determine c ; it merely corrects the finite-range behaviour. In the limit $x \rightarrow \infty$ the exponential factor tends to e^c , which could be any positive number. The data up to 10^{19} give $G(x)/(\log x)^2 \approx 0.8\text{--}0.9$, suggesting $e^c \approx 0.85$ if the asymptote has already nearly been reached.

The distribution of $F(x)$ is intimately linked to the spacing of zeros. Montgomery's pair correlation conjecture implies that the zeros behave like a Gaussian process, and the maximum of $|F(y)|$ over $y \leq x$ would then be of order $\log x \cdot \sqrt{2 \log \log x}$. This would make $\log Z(x) \sim \log \log x + \frac{1}{2} \log \log \log x$, which still tends to a constant after division by $\log \log x$. Thus the pair correlation conjecture is consistent with the functional form we have discovered, though it does not fix the constant c .

Conclusion

We have presented an empirical formula,

$$G(x) \approx (\log x)^2 \exp\left(\frac{\log Z(x) + 0.4}{\log \log x}\right),$$

that reproduces all known record prime gaps up to 10^{19} with an error of about 12 % on average. The constant 0.4 is obtained from a fit to the data; its theoretical origin remains an open problem, possibly related to the pair correlation of zeta zeros. The formula is asymptotically equivalent to Cramér's conjecture if the exponential factor tends to a constant (which our data suggests is near 0.85 for the computed range).

We make no claim of having proved any of the classical conjectures. Instead, we offer this relation as a testable heuristic that may guide further numerical work and inspire new analytical approaches. All code and data are available at <https://github.com/Kappian8858/prime-gap-zeta-zeros>.

99 Cramér, H. (1936). On the order of magnitude of the difference between consecutive prime numbers. *Acta Arithmetica*, 2, 23–46. Davenport, H. (1980). *Multiplicative Number Theory*. Springer. Montgomery, H. L. (1973). The pair correlation of zeros of the zeta function. *Proc. Symp. Pure Math.*, 24, 181–193. OEIS Foundation Inc. (2026). A002386 (lower of record gaps) and A005250 (gap sizes). The On-Line Encyclopedia of Integer Sequences. Kappian KT. (2026). *prime-gap-zeta-zeros*: Python code and data for the empirical formula. GitHub repository, <https://github.com/Kappian8858/prime-gap-zeta-zeros>.